

BHANU NAGPURE

Nagpur, MH, India | +91 982-210-7989

bhanunagpure453@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Summary

Flutter & Python engineer with global research internship experience, a live production app (PyMaster) available in 170+ countries, two international research publications, and a granted patent. Passionate about building scalable mobile and AI-powered solutions.

Education

G H Raison College of Engineering

Bachelor of Technology in Computer Science and Engineering

CGPA: **8.7 / 10.0**

Courses: DBMS, Web Development, OOPs, Data Structures & Algorithms, Computer Networks

Nagpur, MH, India

December 2021 – June 2025

Skills

- **Programming:** Python, Kotlin, Java, Dart, C++, C, SQL, JavaScript, HTML, CSS
- **Frameworks:** Flutter, TensorFlow, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy
- **Tools:** Git, GitHub, REST APIs, Google Cloud, Android SDK, Firebase, Riverpod 2.0
- **Methodologies:** ML, Deep Learning, Mobile App Development, IoT, UI/UX, Agile, Project Management

Professional Experience

Founder & Lead Mobile Engineer

Devanshu Studios

August 2025 – Present

Remote

- Architected and shipped **PyMaster** (live on Play Store) – a production-grade Python learning platform (50k+ LOC) using Flutter and Clean Architecture, available in 170+ countries with a 5.0 rating.
- Engineered a custom Native AI Tutor using Gemini LLM for context-aware debugging assistance.
- Built an offline-first sync engine with batched Firebase writes, ensuring zero data loss during network interruptions.
- Implemented scalable state management (Riverpod 2.0) and monetization via RevenueCat (IAP) and Google Mobile Ads.

Machine Learning Intern

Athabasca University

August 2024 – October 2024

Athabasca, Alberta, Canada

- Delivered **95% prediction accuracy** for a text ideology identification model using Transformers/BERT.
- Developed and deployed a prototype web interface for model access, enabling researchers to validate outputs.

Globalink Research Intern

Mitacs

April 2024 – August 2024

Edmonton, Alberta, Canada

- Led end-to-end development of a cross-platform (iOS, Android, Web) field assignment app using Flutter, handling UI/UX, frontend and backend implementation.
- Presented research at the **3rd Annual TRESL Lab International Symposium**, Athabasca, Canada.

Software Developer Intern (BETiC)

BETIC-GHRCE

October 2022 – May 2024

Nagpur, MH, India

- Engineered an Android app in Kotlin for TMJ disorder screening, enhancing classification accuracy by 70%.
- Contributed to a **granted patent** (No. 579268) for the Smart Chairside TMJ Examination Tool.

Fellowship

Indian Institute of Technology Bombay

March 2022 – October 2022

Mumbai, MH, India

- Led a team to design and prototype solutions for xerostomia patients; received commendation from IIT Bombay industry experts.
- Won the **Most Ergonomic Device Award** at MEDIC 2022 for the Xerostomia Estimator Device.

Projects

V-Safe-Anywhere: Wearable AI & IoT Safety Device

Flutter, Firebase, IoT, Google Apps Script

November 2022 – April 2024

- Developed IoT-based wearable with real-time Firebase sync, emergency alerts & 98% uptime.
- Published research paper in **IGI Global** (March 2024).

Smart Device for Early Screening of OSMF

Android, IoT, Firebase, Python

December 2022 – July 2023

- Built OCanPredict Android app + biomedical sEMG device, improving diagnostic efficiency by 40%.
- Received **multiple national and state-level awards**; project selected for technology transfer.
- Secured **Rs. 10 Lakh BIRAC E-YUVA Government Grant** for further development.

Autonomous Floating Garbage Collection Device

June 2023 – September 2023

Python, YOLOv5, NumPy, IoT, ESP32-CAM, Arduino

- Designed a computer vision system reducing manual waste collection by 60%; open-sourced on GitHub.
- Published in **ACM** and presented at the **8th International Conference on Algorithms, Computing and Systems**, Hong Kong (October 2024).

Open Source Contributions

Wikimedia Foundation – Scribe-Android

February 2025 – Present

- Implemented tablet-specific keyboard layouts for 7+ languages (German, French, Spanish, etc.).
- Fixed UI bugs related to animations, theme consistency, and emoji handling; refactored KeyHandler module.

Publications

1. V-Safe-Anywhere: Empowering Women’s Safety with Wearable AI and IoT

IGI Global

March 2024

2. Autonomous Floating Garbage Collection Device using Computer Vision and Robotics

Association for Computing Machinery (ACM), ICACS 2024

February 2025

Patent

Smart Chairside TMJ Examination and Muscle Activity Detector Tool

Patent No. 579268 | Application No. 202321022476

Granted, March 2023

Achievements

- **Winner** – SPIRIT '25 Health-Tech & AI Hackathon, IIT BHU Varanasi March 2025
- **1st Place** – TechTrek Hackathon 2025, Nagpur January 2025
- **3rd Rank (State)** – 16th Avishkar Inter-University Research Convention, MUHS Nashik January 2024
- **Runner-up** – SRISHTI 2023, Saintgits College of Engineering, Kerala March 2023
- **Runner-up** – DIPEX 2023, Amravati April 2023
- **Rs. 10 Lakh Grant** – BIRAC E-YUVA, Government of India February 2023
- **Winner** – MEDHA & MEDIC 2022, IIT Bombay October 2022

Conferences

- **3rd Annual TRESL Lab International Symposium**, Athabasca, Canada July 2024
Presented: “Development of A Mobile Field Trip Application”
- **8th International Conference on Algorithms, Computing and Systems (ICACS 2024)** October 2024
Hong Kong Metropolitan University, Hong Kong – Presented autonomous garbage collection device research

Positions of Responsibility

Student Nominee – Interdisciplinary Board of Studies (IBoS)

December 2022 – December 2024

G H Raisoni College of Engineering, Nagpur

Represented students in curriculum reform discussions under NEP-2020.

Volunteer Experience

Oral Cancer Awareness Camps

December 2022 – June 2023

- Educated over 1,000 individuals on tobacco and betel nut risks in collaboration with Lata Mangeshkar Hospital.
- Trained primary healthcare workers in rural areas to operate the OSMF screening device.

Certifications

- Building Transformer-Based NLP Applications – **Nvidia**
- Postman API Fundamentals Student Expert – **Postman**
- Medical Device Innovation Course – **IIT Bombay**

References

- Dr. Qing Tan, Athabasca University – qingt@athabascau.ca
- Dr. Vibha Bora, G. H. Raisoni College of Engineering – vibha.bora@raisoni.net
- Dr. Snehlata Dongre, G. H. Raisoni College of Engineering – snehlata.dongre@raisoni.net